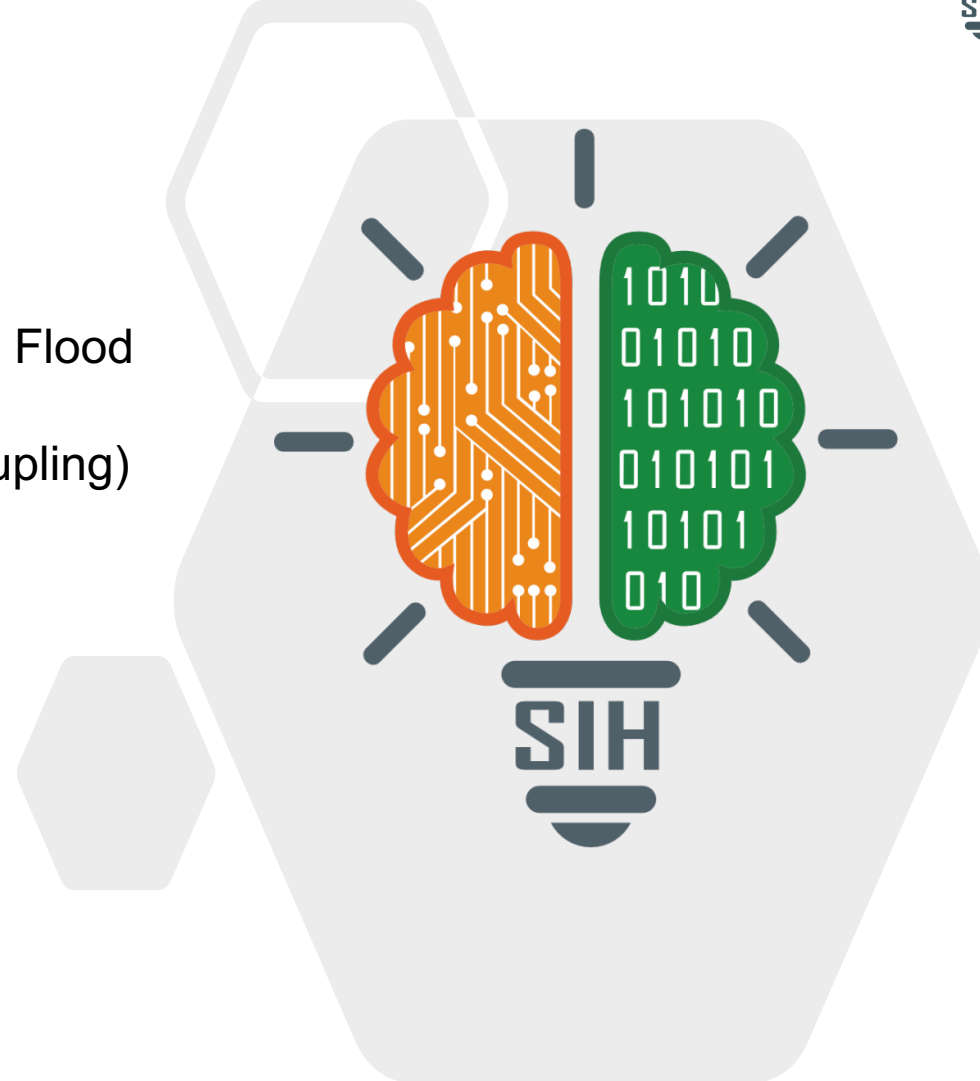


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26085
- **Problem Statement Title-** Urban Flood
Nowcasting System (Drainage and Rainfall Coupling)
- **Theme-**
- **PS Category-** Software
- **Team ID-**
- **Team Name** - Hexaura



The Challenge

Urban flooding is an annual crisis in Indian cities. Weather forecasts alone cannot predict where streets will flood. Fragmented citizen reports, CCTV, drainage, and historical data make it difficult to identify flood locations, severity, and vulnerable areas—delaying emergency response.



Proposed Solution

FLOODSHIELD AI combines **citizen reports, CCTV, drainage data, historical floods and weather data** to analyze flood hotspots and vulnerable streets, delivering **street-level predictions, early warnings and safe routes** for faster response.



DATA → INSIGHTS → EARLY ACTION → RESILIENT CITIES

- IMPACT AT A GLANCE (INDIA) -

54,473
urban local bodies

100+
million urban citizens

Safer &
more resilient cities

KEY BENEFITS



SUPPORTED CITIES (CONFIGURABLE)



EARLY INSIGHTS. FASTER ACTIONS. FLOOD RESILIENT INDIA.

01 DATA SOURCES (No Physical Sensors Required)

-  **Radar / Rainfall Datasets**
(IMD, GPM, INSAT)
-  **Weather APIs**
(Open-Meteo, IMD)
-  **High-resolution DEM**
(SRTM / CartoSAT)
-  **GIS Road Network**
(OpenStreetMap)
-  **Municipal Drainage GIS**
(Open data / city datasets)
-  **Land Use / Land Cover**
(Imperviousness, NLCD)
-  **Historical Rainfall & Flood Records**
-  **CCTV Images & Videos**
(City feeds, public sources)
-  **Citizen Flood Reports**
(Photo + Location + Depth)
-  **Deterministic Simulated Storm**
(For demo / scenario testing)

Every value labelled: LIVE | HISTORICAL | SIMULATED | DEMO

02 PREDICTION ENGINE (AI + Physics Hybrid)



03 DECISION OUTPUTS (From Prediction to Action)

-  Street-level flood map
(depth, extent, risk zones)
-  Time-to-flooding & duration
-  Flood propagation animation
-  Alert levels
(Green / Yellow / Orange / Red)
-  Flood-safe emergency routes
-  Hospital & infrastructure impact
-  What-if scenario comparison
-  Drainage maintenance priorities
-  PDF incident report
-  Grounded AI Copilot answers
(for officials & citizens)

04 PLATFORM & DELIVERY (For Citizens, Authorities & Responders)



TECH STACK





Technological Feasibility

Built entirely with existing software technologies: GIS, AI/ML, computer vision, rainfall APIs, DEM data, PostGIS, hydraulic modelling and cloud infrastructure. **No physical sensors or IoT hardware required.**



Data Feasibility

Combines citizen reports, CCTV images/video, drainage infrastructure, historical flood records, weather/rainfall data, DEM and road-network datasets. Real, historical, simulated and predicted data are **clearly separated.**



Operational Feasibility

A unified command-center platform gives municipal officers street-level flood intelligence, alerts, emergency routing, infrastructure impact and what-if simulations from **one interface.**



Integration Challenges and Solutions

Different data formats, spatial resolutions and update frequencies are handled through APIs, geospatial preprocessing, validation and a **unified PostGIS data layer.**



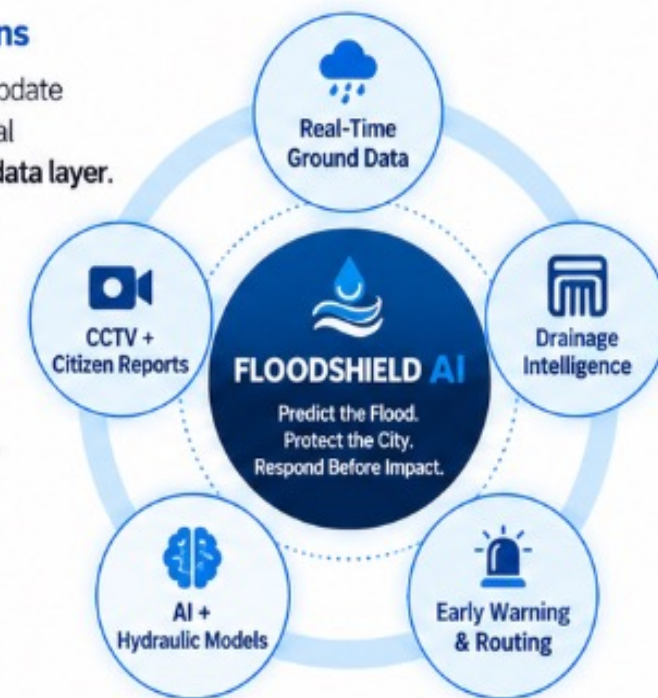
Adoption Strategy

Start with configurable city deployments, integrate existing municipal datasets and workflows, and provide **role-based dashboards** for officers, responders, analysts and citizens.



Scalability

The architecture is **city-agnostic** and can expand from one pilot city to multiple Indian cities using the same software platform and configurable datasets.



DATA SOURCES

Citizen + CCTV + Weather + DEM + Drainage + Historical Floods



CORE ENGINE

GIS + AI/ML + Computer Vision + Hydraulic Analysis



DECISION SUPPORT

Flood Prediction + Risk + Alerts + Routing + Simulation



IMPACT

Faster Response + Safer Routes + Better Urban Planning



Saves Lives
Early warnings help citizens and emergency services act in time.



Reduces Economic Losses
Minimizes damage to property, businesses and infrastructure.



Builds Climate Resilient Cities
Helps cities adapt to extreme weather events.



Supports Sustainable Planning
Data-driven decisions for long-term resilience.



Empowers Communities
Real-time information and safer routes for everyone.





Faster Response
Real-time alerts and actionable insights.



Better Decision Making
Accurate and data-driven predictions.



Cost Efficient
Uses existing data sources, no new hardware required.



Scalable & Replicable
Can be deployed in any Indian city.



Inclusive
Benefits authorities, responders and citizens alike.



-  Late Warnings
-  Blocked Roads
-  High Damages



-  Early Warnings
-  Safe Routes
-  Reduced Losses



KEY IMPACT METRICS



50-70%
Flood-related casualties



30-60%
Economic losses



3-6 hours
Advance warning time



10M+
Citizens benefited across Indian cities



**CLEANER CITIES
SAFER COMMUNITIES
STRONGER INDIA**



Live Demo



<https://floodshield-web.onrender.com/login>



Scan to open



GitHub Repository



<https://github.com/bayyavarapubhanuteja/floodshield-ai.git>



Scan to open

 SOURCE	 TITLE	 SIGNIFICANCE
Environmental Modelling & Software Vol 143, September 2021 ScienceDirect.com by Elsevier	Environmental Modelling & Software	Supports using webcams, social media and citizen-science data for fine-scale urban flood monitoring.
Sustainable Cities and Society Vol 137, 1 February 2026 ScienceDirect.com by Elsevier	Sustainable Cities and Society	Supports combining ML and geospatial/SAR data for urban flood-risk mapping and infrastructure planning.
SOP-Urban-flooding.pdf	Urban Flooding During Monsoon Season	Provides the Indian government context on urban drainage, flood mitigation and the role of Urban Local Bodies
Urban Climate Vol 53, January 2024 ScienceDirect.com by Elsevier	Floods recent years with crowdsourced data	Directly supports your citizen/crowdsourced flood-data approach and its combination with rainfall and elevation information.